

Review of ‘Phylogenomic assessment of microhylid frogs reveals widespread taxonomic confusion in the Asterophryinae and establishes the timing of diversification in Australia’ by Brennan et al.

This is a joint review by a PhD student and their supervisor.

Summary: This study describes a novel phylogenomic dataset (anchored hybrid enrichment) for microhylid frogs (Anura: Microhylidae), a family with several unresolved higher-level relationships. It has a special focus on Asterophryinae and understanding the origins of the Australian members of this subfamily. Overall, we thought the manuscript was well-written and concise. Strengths of the manuscript include the figures (which are exceptional), a clear introduction into microhylid systematics, and a novel and interesting phylogenomic dataset. While we found the paper to be noteworthy and publishable, we also identified several areas for improvement which we describe below.

General comments:

Morphological analyses: We identified two main issues with the morphological data and analyses as currently presented. First, as described in the text, the data are taken from previous studies, yet the authors did not address data compatibility across these datasets. Interobserver variation in character measurement is a known issue in frog morphometrics (see Hayek et al. 2001, *Alytes*, Frog morphometrics: a cautionary tale). It also is not clear exactly how the measurements were taken. For example, what landmarks were used to define head width (this differs across some anuran studies)? A table with summary statistics for the measurements along with references would be helpful and improve interpretability of the morphological results. This could be a supplementary table. Second, many of the diagnostic features that have been used to develop the taxonomy of Asterophryinae are based on soft or internal anatomy (e.g. vertebral column morphology or tongue morphology). These characters are not considered as part of the morphometric analysis presented here, yet the authors seem to conclude that morphology is (in general) quite bad at delimiting diversity based on their analysis. Many of the eight morphological measurements they use are highly conserved across anurans or associated with ecological habitats and not taxonomy (e.g. eye diameter; Thomas et al. 2020, *Proc. Roy. Soc. B*). We encourage the authors to either revise their text or include a broader comparative examination of morphology. Here is one of the relevant (and currently uncited) studies that may help with a deeper examination of asterophryine morphology:

Burton, T. C. 1986: A reassessment of the Papuan subfamily Asterophryinae (Anura: Microhylidae). Records of the South Australian Museum 19:405–450.

Missing citation: Portik et al. (2023, Mol. Phylogenet. Evol.) also discuss microhylid phylogenomics and issues within Asterophryinae. This study features several similar results that seem relevant for the discussion provided in this manuscript (e.g. placement of *Aphantophryne pansa* with *Cophixalus* and a well-supported Microhylinae + Dyscophinae).

Figure 1 data: The species richness and geographic patterns of diversity are not described in the text. Where these data come from and how they were analysed should be discussed in the text.

Data availability: Are the phylogenomic and morphological data available? There is no mention of GenBank SRA or where the morphology matrix may be obtained.

Specific comments by line number:

Line 57: Confusing use of ‘ranging’. We think it would be clearer if the others used ‘occur’.

Line 62: I am not sure that I would agree that the consensus is for most subfamilies (i.e. intrafamilial relationships). The cited studies vary in their placement of several subfamilies and for some subfamilies (e.g. Melanobatrachinae) there are no phylogenomic data.

Line 80-81: Confusing sentence, should be something like, ‘...represent a single colonization event...’

Line 174: What 40 publications exactly? These should be listed either in text or as an Appendix.

Line 180: Additional random forest analysis details are required. For example, the number of trees.

Line 184: What criteria were used to determine that only the first two axes should be examined in plots for the PCA, LDA, and FDA?

Line 205: 'Tree topology' is redundant. Just use one.

Line 218: If *Mantophryne* was not sampled, how can you comment on its placement? From the mitochondrial analysis with the Hill et al. (2022) data?

Line 234: How many species were represented in each of the genera that had their morphology compared?

Line 239 / Figure S4: When genera were misclassified, which incorrect groups did the RandomForest method, LDA, and FDA place them with?

Line 264: How can you differentiate between elevated Cretaceous extinction versus low lineage birth during the Cretaceous?

Line 317: Missing multiple key morphological studies at the genus level (e.g. Burton study mentioned previously).

Line 335: There are no *Mantophryne* sampled in the main phylogenomic study, so please be clear about what is meant here.

Line 408: *Cophixalus* is labelled as "top" and *Austrochaperina* is labelled as "bottom," but they are both on the bottom, located left and right of each other respectively. Additionally, in all other instances the location precedes what is described while this line has the location afterwards. For consistency, something could be written like, "... species richness in (right) *Cophixalus* and..."

Line 484: Note tab formatting issue with references (Hill and Hime references).

Figure 1: Where do these species richness data come from? Not addressed at all in the text.

Figure 3: Datapoints appear 'ghostly' on some computer screens. The main data points should be darker so that the entire dataset is visible to a wide range of readers. Also, some of the frog silhouettes are covering data points and obscure the information being presented.